

Norm of V a vector

= length of V

Pythagorean The



$\|V\|$ = Norm of V

$$\|V\|^2 = \sum_{i=1}^n V_i^2$$

length

$$= \sqrt{x^2 + y^2}$$

$$= \sqrt{x^2 + y^2 + z^2}$$

In data + statistics

total squared error (TSE)

error = actual - true

if its a vector = V

$$TSE = \sum_{i=1}^n V_i^2$$

$$\text{Mean Squared Error} = \frac{TSE}{n}$$

$$TSE = \|V\|^2$$

$$\begin{array}{c} \uparrow \\ \text{Scalar} \end{array} 6 \cdot \begin{pmatrix} 3/2 \\ 4/3 \end{pmatrix} \overset{\text{Component wise}}{=} \begin{pmatrix} 6 \cdot 3/2 \\ 6 \cdot 4/3 \end{pmatrix} = \begin{pmatrix} 9 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} 3/2 \\ 4/3 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3/2 - 1 \\ 4/3 + 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 7/3 \end{pmatrix}$$

$$A v = u$$

$$u_i = \sum_{k=1}^n a_{ik} v_k$$

$$A v = \lambda v$$

eigen vector
eigen vector
equation

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = u, \quad u_i = \lambda v_i$$

Def: A is diagonal
if $\forall i, j$ with $i \neq j$
 $a_{ij} = 0$

Off diagonal elts are 0

Fact: eigenvalues of a diagonal
are the entries on diagonal

Note: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is a diagonal
matrix

Identity Matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$
 $a_{ii} = 1$

$$I \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \cdot x + 0 \cdot y \\ 0 \cdot x + 1 \cdot y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Zero Matrix = 0

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad a_{ij} = 0$$

$$0 \cdot v = 0 \text{ vector}$$

$$I \cdot v = v \quad \forall v$$

$$\sum a_{ik} v_k = a_{kk} v_k$$

$$2 \cdot \frac{1}{2} = 1 \quad \frac{2}{3} \cdot \frac{3}{2} = 1$$

$$(\frac{3}{2}) = (\frac{2}{3})^{-1} \text{ mult inverse}$$

$$A \cdot B = I \quad A, B \text{ } n \times n \text{ matrices}$$

Square matrices

$$\text{then } B = A^{-1}$$

Fact: In general $AB \neq BA$
but if $AB = I$ then $BA = I$

inverse equation. Rows = Cols

Recall: $F: X \rightarrow Y$ is invertible
iff F is 1-1
and onto

$$X = \{-1, 0, 1\} \quad F(x) = x^2$$

	x^2	
-1		1
0		0
1		1

this F is not 1-1, onto

IF F is invertible

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$a_{11} = 2 \quad a_{12} = 0$$

$$~~a_{12} = 0~~$$

$$a_{21} = 0 \quad a_{22} = 3$$

$$v = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$Av = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$a_{11}v_1 = 2x$$

$$a_{12}v_2 = 0 \cdot y$$

$$a_{21}v_1 = 0 \cdot x$$

$$a_{22}v_2 = 3 \cdot y$$

$$= \begin{pmatrix} 2x + 0y \\ 0x + 3y \end{pmatrix}$$

$$= \begin{pmatrix} 2x \\ 3y \end{pmatrix}$$

$$Av = \lambda v$$

$$\lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

$$\begin{pmatrix} 2x \\ 3y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

Both

$$2x = \lambda x$$

$$(\lambda - 2)x = 0$$

$$3y = \lambda y$$

$$x = 0 \wedge \lambda = 3$$

$$y = 0 \wedge \lambda = 2$$

$$x = 0 \wedge y = 0$$

eigenvector
- eigenvalue
 $x \neq 0 \vee y \neq 0$

$$\lambda = 2 \wedge y = 0$$

$$\lambda = 2 \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix} \quad x \neq 0$$

↑

$$\text{Pick } x = 1 \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Eigenvalue

Eigenvector

$$\lambda = 3 \wedge x = 0$$

↑

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ y \end{pmatrix} \quad y \neq 0$$

Eigenvalue

$$\text{Pick } y = 1 \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ eigenvector}$$

eigenvalues and eigenvectors

$$\text{of } A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

x	F
-1	1
0	0
1	-1

IF F is invertible

Size of X

= Size of Y

For Matrices or linear maps

dimension of X

= dimension of Y

When $F(v) = Av$
is invertible

When A^{-1} exists
 $\Rightarrow$ rows = cols

Dimension = # entries of vector
size of vector
Shape
= # of free variables

Singular Value Equation

$$A v_i = s u_i$$

s scalar $s \geq 0$ always real

if $s=0$
then v is in Nullspace

A can be rectangular

Null space of A

$0 \in \text{Nullspace}$

Solutions to $Av = 0$

$$A = \begin{pmatrix} 1 & 1 \end{pmatrix}_{1 \times 2} \quad x+y=0 \quad \text{Sol: } \text{Span} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \left\{ \begin{pmatrix} x \\ -x \end{pmatrix} \mid x \in \mathbb{R} \right\}$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{2 \times 3} \quad \begin{matrix} x+y=0 \\ z=0 \end{matrix} \quad \text{Sol: } \text{Span} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \left\{ \begin{pmatrix} x \\ -x \\ 0 \end{pmatrix} \mid x \in \mathbb{R} \right\}$$

Expect nontrivial Nullspace

Cols $>$ row